

DEPARTMENT OF MINES
SOUTH AUSTRALIA

GEOLOGICAL SURVEY
REGIONAL SURVEYS DIVISION

THE ADELAIDE GEOSYNCLINE AND STUART SHELF

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NOTE from Author

The text of this report may differ in some minor details from the finally edited version that is in publication in Economic Geology of Australia and Papua New Guinea.

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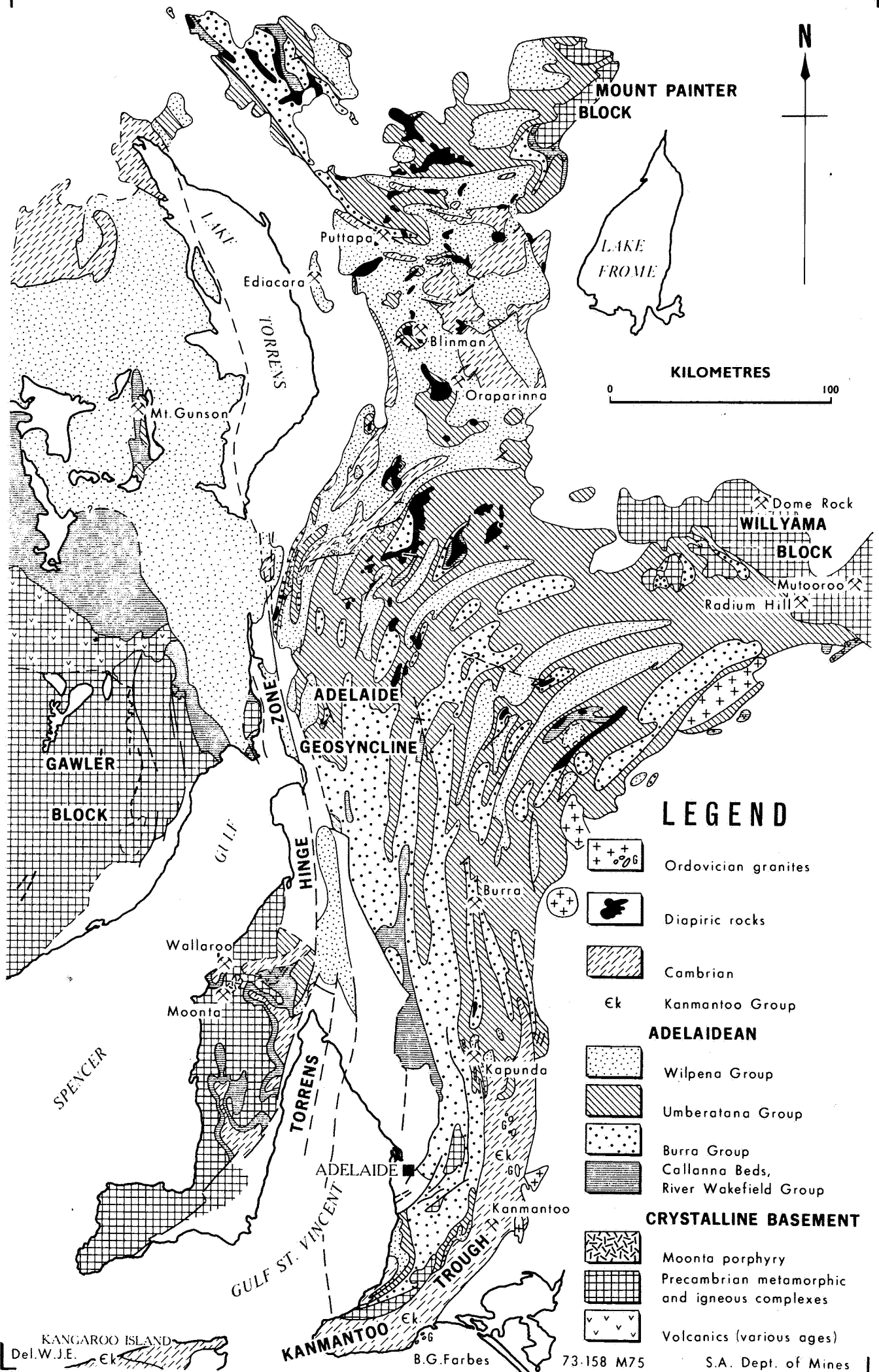


Fig. 1. GEOLOGICAL MAP OF ADELAIDE GEOSYNCLINE

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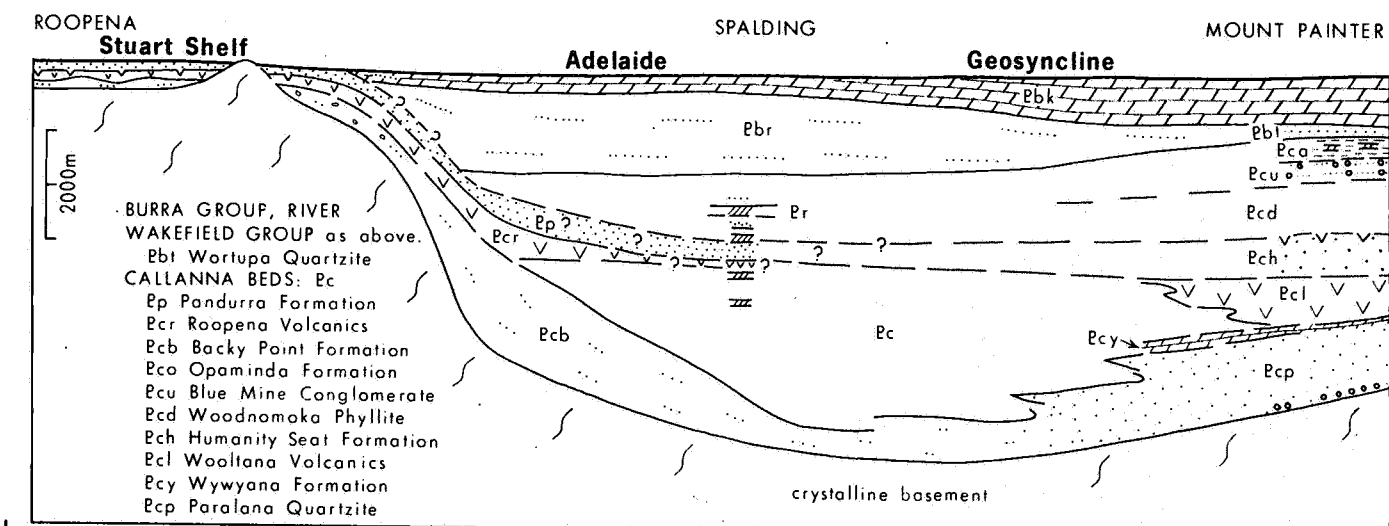
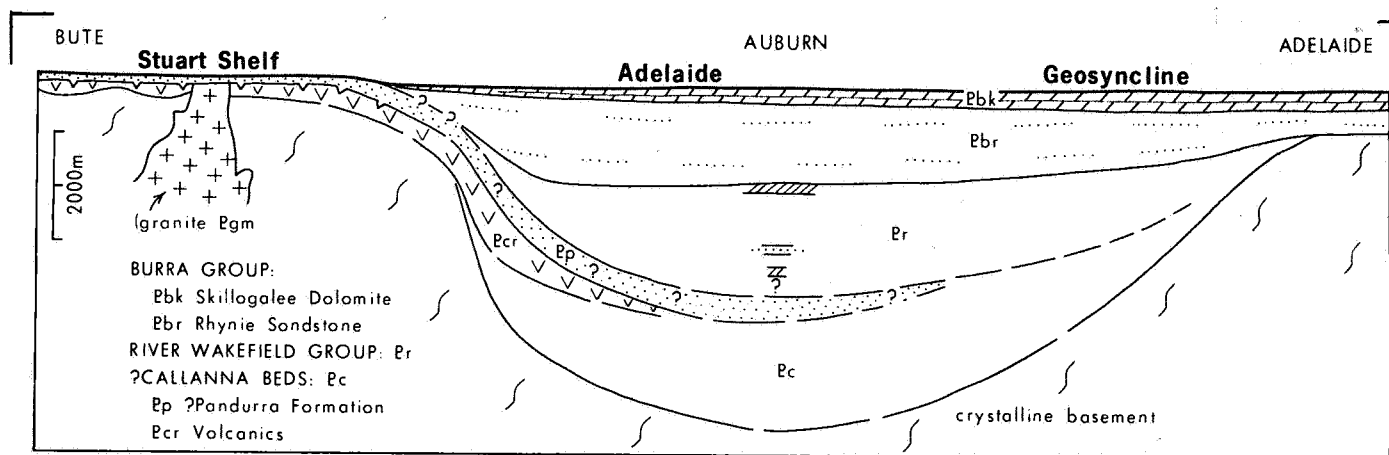
INTRODUCTION

An arcuate belt of Precambrian sediments which overlies unconformably older crystalline basement extends northerly and westerly across South Australia. The sediments represent the cover of a former more extensive platform. The most complete record of these and overlying Cambrian sequences is preserved in the Adelaide Geosyncline (Sprigg, 1952). The Geosyncline is an important metallogenic province and in this paper is described as extending from the Kangaroo Island - Mount Lofty Ranges area near Adelaide to the Willouran Ranges near Marree in the north.

The time term Adelaidean is applied to the Precambrian sequences preserved in the Geosyncline. The following table outlines the scheme of Adelaidean time subdivisions and rock groupings adopted by the Geological Survey to describe these sequences.

TABLE I
Subdivisions of Adelaidean Sequences

PRECAMBRIAN TIME TERMS	ROCK GROUPINGS
MARINOAN	WILPENA GROUP
	(glaciation)
STURTIAN	UMBERATANA GROUP
	(Glaciation and tectonism)
TORRENSIAN	BURRA GROUP



M75 Del.C.R.S. 73-225

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Fig. 2. DIAGRAMMATIC SECTIONS; BUTE - ADELAIDE AND ROOPENA - MT. PAINTER

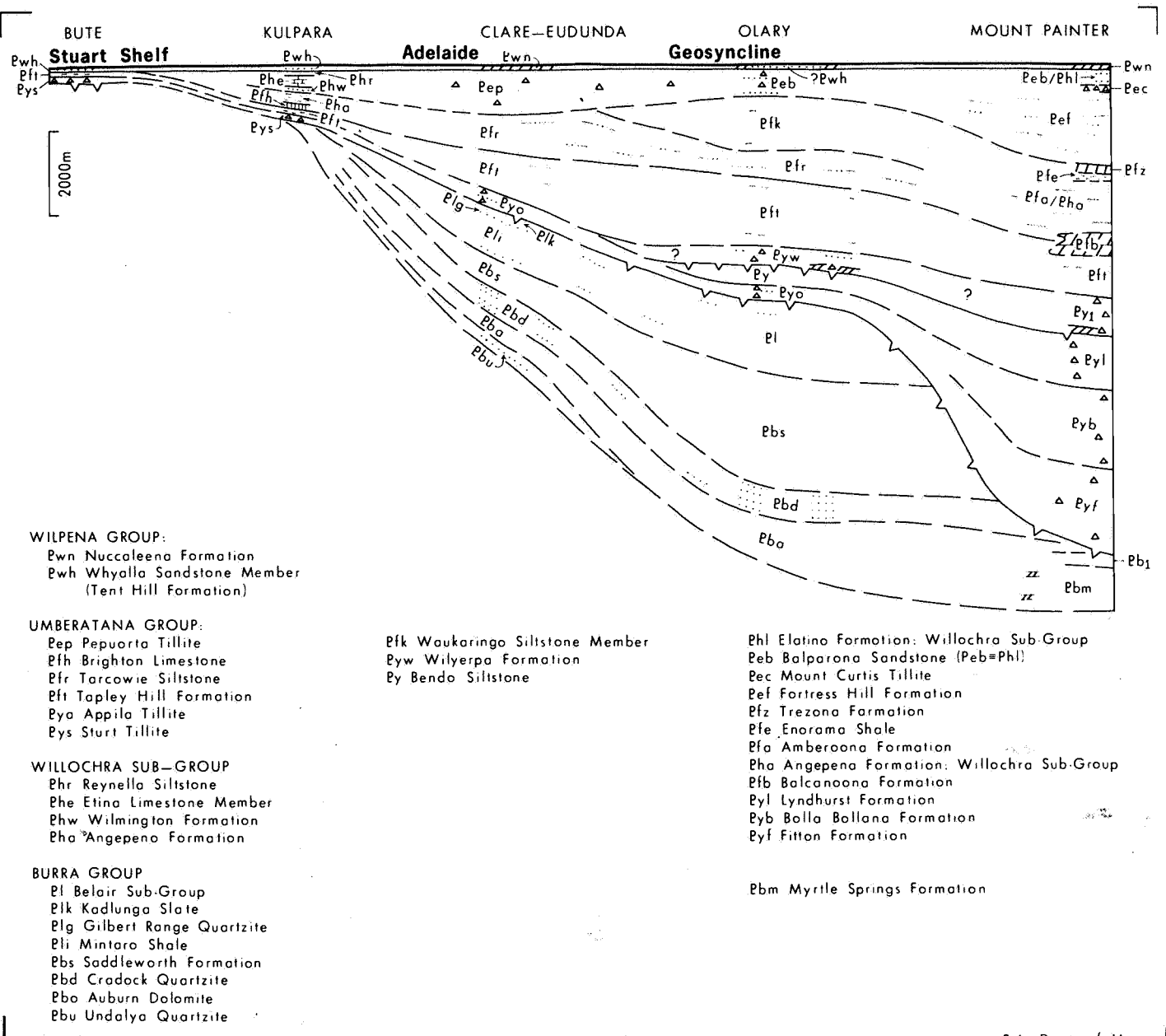
WILLOURAN	UPPER CALLANNA BEDS
	(?tectonism) LOWER CALLANNA BEDS

The largely undeformed and relatively thin Adelaidean remnant cover on the Gawler Craton to the west of the Geosyncline defined the Stuart Shelf. The complex transition zone between Shelf and Geosyncline is the Torrens Hinge Zone, a near-meridional feature that forms part of a continental fracture pattern of major tectonic and metallogenic significance.

For details of stratigraphic sequences readers are referred to the articles listed in the short bibliography, and in particular, the Handbook of South Australian Geology (L.W. Parkin, Ed. 1969). Below are a few comments on each of the main rock groups.

In the Mount Painter region the Callanna Beds rest unconformably on crystalline basement about 1 600 m.y. old (Carpentarian): they form a thick sequence both here and in the Willouran Ranges. The lower Callanna Beds are separated from the upper Callanna Beds by an erosional break and contain the Wooltana Volcanics which are possibly to be correlated with unnamed volcanics in the Peake and Denison Ranges, the Wilangee Volcanics in western New South Wales (Rose and Brunker, 1971, p. 111) and blocks in diapirs. The position of the River Wakefield "Group" in relation to the upper Callanna Beds (with which it was previously correlated) is now in doubt, since recent work by W.V. Preiss (personal communication, 1973) suggests that it lies above the Rhynie Sandstone in the Burra Group.

The Burra Group seems to represent continuation of the shallow water conditions in which the Callanna Beds were deposited and contains in the Skillogalee Dolomite, sedimentary magnesite beds, stromatolites (Preiss, 1971) and fossil algal filaments (Schopf and Barghoorn, 1969).



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Fig. 3. DIAGRAMMATIC SECTION; BUTE - OLARY - MT. PAINTER

The Umberatana Group is characterised by tillitic rocks representing three phases of glaciation. The Braemar Iron Formation, possibly of lacustrine origin, lies between the two earliest phases. The first and second phases of glaciation were preceded by tectonism.

The Wilpena Group features thick sequences of silty and finely sandy rocks and contains near its top the Ediacara Fauna. The overlying Cambrian has been described by Daily (1972).

The Lower Cambrian is rich in carbonate rocks (Daily, 1969) and records a marine transgression. In the Flinders Ranges the Lower Cambrian Hawker Group was succeeded after a slight hiatus (disconformable base of the Billy Creek Formation) by red beds of the Middle to ?Upper Cambrian Lake Frome Group.

TECTONIC EVOLUTION OF THE ADELAIDE GEOSYNCLINE

Tectonic Setting

The Adelaide Geosyncline is flanked on the west and east by the Gawler Craton and Willyama and Mount Painter Inliers comprising Precambrian crystalline basement complexes radiometrically dated as older than 1 400 m.y. The term Geosyncline is applied because of the great thickness, in excess of 24 000 metres (80 000 ft.) represented by the Adelaidean and Cambrian sediments within this framework. The Geosyncline marks an ancient region of platform downwarp bounded by marginal rifts. The basement underwent intervals of tectonism followed by quiescent periods of gentle subsidence. The sequences were deformed in the Adelaidean and in the Cambro-Ordovician. The sediments preserved are generally of continental-lagoonal-shallow marine facies mainly derived from the crystalline basement in the west. The Geosyncline was intermittently connected to other basins to the north and east.

Early Willouran Environments and Moonta - Wallaroo Mineralisation and
Middle Willouran Tectonism

Flood basalts, of the Wooltana Volcanics type with minor copper mineralisation, were a feature of the early Adelaidean. They probably reflect tensional rifting and are associated with carbonate sedimentation and evaporities. The dating of the earliest Adelaidean is dependent on the correlation between the Wooltana and the Roopena Volcanics (1 345 m.y.) on the eastern and western flanks of the Geosyncline. Volcanics and metasiltstones in drill holes in the Willamulka - Bute area east of Moonta - Wallaroo district are either correlated with the Roopena Volcanics or an older sequence. These rocks were deformed and metasomatised, presumably by the 1 200 m.y. event* recorded radio-metrically by A.W. Webb (pers. comm.) in granites in the Moonta region. This event probably records the emplacement of the Moonta - Wallaroo copper lodes. The carbonate metasomatism, early deformation and possible igneous intrusion of the lower Callanna Beds in the Geosyncline may have been initiated at about the time of the Moonta mineralisation event in structures which were to evolve into centres of diapirism. This activity may reflect basement movements associated with the Musgravian Orogeny.

Late Willouran and Torrensian Environments

The thick sequences of the continental and shallow marine upper Callanna Beds, River Wakefield Group and Burra Group and an unknown proportion of the Duff Creek Formation to the north and the Pandurra Formation to the west followed the earlier Willouran tectonism.

*Subsequent to preparation of the ms for publication Dr. Webb advises that further Rb/Sr and K/Ar geochronological data indicates that the granite age is more likely to be c. 1 500 m.y. and therefore probably earlier than the lower Callanna Beds (B.P.T.)

The Sturtian - Marinoan Interval

Uplift, diapirism and graben development in the north marked the onset of the earliest Sturtian glaciation. The subsequent interglacial interval was probably associated with an eustatic rise in sea level and climatic change. The Sturtian unconformity is a locus for later copper mineralisation at the base of the Woocalla Dolomite, as is the contact between Sturtian glacials and pyritic Tindelpina Shale Member in the Geosyncline. Following the Marinoan glaciation, there was again transgression onto the Stuart Shelf. This Marinoan transgression is a locus for the Mount Gunson stratiform copper mineralisation in the Whyalla Sandstone Member.

Cambrian Sedimentation, Tectonism and the Delamerian Orogeny

The shallow water Marinoan sedimentation of largely western provenance, was terminated by uplift over much of the Geosyncline. The onset of the Cambrian was marked by a change to an eastern provenance and a widespread change to a marine shelf carbonate environment. Early Cambrian volcanism occurred in the south. Geochemical evidence (Thomson 1965) indicates that lead, copper and zinc enrichments were widespread at the transition zone and associated with Mississippi Valley type ore occurrences (e.g. Ediacara lead). Late Lower Cambrian positive movements followed. Following a brief return to marine carbonate deposition in the Middle Cambrian, shallow water and continental sedimentation terminating probably in the Upper Cambrian, marked the onset of the Delamerian Orogeny (Thomson, 1969). The orogeny transformed the Adelaide Geosyncline into a fold belt and concluded with the intrusion of Ordovician granite (about 450 m.y.) in the east and south, e.g. at Anabama and Bendigo (S.A.) associated with low grade copper mineralisation.

The regional metamorphic gradient across the Geosyncline led to an extensive reheating of the Geosyncline basement which was probably already mineralised and thus provided a secondary source of metal by fluid transfer.

Diapirism, carbonate intrusion and metasomatism was particularly active at basement structural intersections which were also the loci for basic and probably acid intrusion, e.g. porphyry at Burra. It is likely that the majority of the base metal deposits in the Geosyncline were finally localised in their particular stratigraphic or structural settings as a result of the Delamerian Orogeny. Large scale movements of intrastratal fluids containing traces of base metals commenced during diagenesis and with increasing pressure and temperature moved upwards around diapirs and westward to the Torrens Hinge Zone where wedge-outs and disconformities between favourable facies became the sites for stratiform base metal mineralisation.

REFERENCES

- Coats, R.P., 1969. Diapirism in the Adelaide Geosyncline. J. Aust. Petrol. Expl. Ass. 1965: 98-102.
- _____ and Blissett, A.H., 1971. Regional and Economic Geology of the Mount Painter Province. Bull. geol. Surv. S. Aust. 43: 426 pp.
- Cooper, P.F. and Tuckwell, K.R., 1971. The upper Precambrian Adelaidean of the Broken Hill area. A new subdivision. Quart. geol. Notes. Geol. Surv. N.S.W. 3: 8-16.
- Daily, B., 1969. Fossiliferous Cambrian sediments and low grade metamorphics, Fleurieu Peninsula, South Australia. Geological Excursions Handbook ANZAAS, 41st Congress, Adelaide: 49-51.
- Daily, B., 1972. The base of the Cambrian and the first Cambrian faunas. Univ. Adelaide Centre for Precambrian Research. Special Paper No. 1 :13-42.
- Dalgarno, C.R., 1954. Report on the Lower Cambrian stratigraphy of the Flinders Ranges, South Australia. Trans. R. Soc. S. Aust. 88: 129-144.
- _____ and Johnson, J.E., 1968. Diapiric structures and late Precambrian - early Cambrian Sedimentation in the Flinders Ranges, South Australia. Mem. Am. Ass. Petrol. Geol. 8: 301-314.
- Forbes, B.G., 1970. Benda Siltstone. Quart. geol. Notes. geol. Surv. S. Aust. No. 33 : 1-2.
- Johns, R.K., 1968. Geology and mineral Resources of the Andamooka - Torrens area. Bull. geol. Surv. S. Aust. 41.
- Preiss, W.V., 1971. Stromatolite biostratigraphy in the Late Precambrian. Quart. Geol. Notes. geol. Surv. S. Aust. No. 38 : 3-11.
- Preiss, W.V., 1973. Palaeoecological interpretations of South Australian Precambrian stromatolites. J. geol. Soc. Aust. 19: 501-532.

- Reyner, M.L., 1955. The Geology of the Peake and Denison region. Rep. Invest. Geol. Surv. S. Aust. 6 : 23 p.
- Rose, G. and Brunker, R.L., 1969. The Upper Proterozoic and Phanerozoic Geology of northwestern New South Wales. Proc. A.I.H. 229: 105-120.
- Schopf, J.W. and Barghoorn, E.S., 1969. Microorganisms from the Late Precambrian of South Australia. J. Paleont. 43 : 111-118.
- Sprigg, R.C., 1952. Sedimentation in the Adelaide Geosyncline and the formation of the continental terrace in Sir Douglas Mawson Anniv. Vol. Ed. M. F. Glaessner and R.C. Sprigg, Univ. Adelaide.
- Thomson, B.P., Coats, R.P., Mirams, R.C., Forbes, B.G., Dalgarno, C.R. and Johnson, J.E., 1964. Precambrian rock groups in the Adelaide Geosyncline : a new subdivision. Quart. geol. Notes. Geol. Surv. S. Aust. No. 9.
- _____ 1965. Source and distribution of heavy metals in Cambrian and Marinoan shelf sediments in South Australia. Proc. (general) V.6. Eighth Comm. Min. Met. Congr. Melbourne. 107-113.
- _____ 1969. The Handbook of South Australian Geology. L.W. Parkin, Ed. Geol. Surv. S. Aust. Ch.1-2, 2-81.
- _____ 1970. A review of the Precambrian and Lower Palaeozoic tectonics of South Australia. Trans. R. Soc. S. Aust. 94 : 193-221.
- Wade, Mary, 1970. The stratigraphic distribution of the Ediacara fauna in Australia. Trans. R. Soc. S. Aust. 94 : 87-104.
- Wilson, A.F., 1952. The Adelaide System as developed in the Riverton-Clare region, Northern Mount Lofty Ranges, South Australia. Trans. R. Soc. S. Aust. 75 : 131-149.
- Wopfner, H., 1970. Lithofacies evaluation of Lower Cambrian sediments of the Flinders Ranges. Mineral Resour. Rev. S.Aust. 129: 11-24.